

JULIAN ZHANG

+1 (647) 914-2970 | julian.zhang@uwaterloo.ca | [linkedin.com/in/zhaju](https://www.linkedin.com/in/zhaju) | zhaju.github.io

EDUCATION

University of Waterloo

(Sep 2024 - Apr 2029)

Bachelor of Computer Science and Finance Double Major (CFM) | *President's Scholarship of Distinction*

EXPERIENCE

RBC Investor Services | Software Engineering Intern

(May 2025 - Aug 2025)

- Automated pension reporting to **500+** Canadian and **100+** international clients using **Python** and **VBA**
- Ensured compliance with service level agreements, achieving **98% accuracy** and **97% timeliness**
- Used **Yahoo Finance** and **scikit-learn** to create a predictive model of when institutional clients buy/sell asset portfolios worth a combined **\$200+ billion**. Presented to upper-level executives

Oakville Centre For Vision | Software Engineer

(May 2024 - Sep 2024)

- Used **C# MonoGame** and **Python** to create a web app for eye-tracking goggles. Used it to analyze vision patterns in professional athletes, collaborating directly with optometrist Dr. Ronald Gall
- Focused on data analysis of **8+ tests** that examine gaze vectors for saccades, fixations, and pursuits
- Collaborated with a hardware team to modernize a legacy codebase for improved functionality

University of Toronto Clean Energy Lab | AI Research Assistant

(Jun 2022 - Sep 2023)

- Applied Crystal Graph Convolutional Neural Network (CGCNN) to examine trends in chemical properties
- Analyzed results for **material discovery** of lithium-ion alternatives for sustainable energy production
- Implemented regression models, achieving an accuracy of **97%**. Presented to Prof. Oleksandr Voznyy

Team Canada Debate | Member

(May 2023 - Aug 2024)

- Represented Canada as **1 of 12** debaters at the 2024 World School Debating Championship (WSDC)
- Conducted research across politics, economics, and international relations, engaging with field experts
- Organized 5+ debate tournaments attracting **500+** participants and raising **\$10,000+** for charity

PROJECTS

SignWave | OpenCV, Flask, MediaPipe, Whisper

- An AI translator that converts text and audio into American Sign Language animations, improving accessibility for the deaf and hard of hearing. **First Place and Best Use of GitHub @ JamHacks VII**

EdgeFund | Yahoo Finance, Pandas, NumPy, matplotlib

- A digital stockbroker statistically selecting a portfolio, outperforming the TSX/S&P by **30% in 1 year**

SKILLS

- Languages:** C/C++/C#, Python, Java, JavaScript, HTML/CSS, Scheme/Racket, LaTeX
- Libraries:** React.js, Node.js, NumPy, matplotlib, Pandas, PyTorch, TensorFlow, Yahoo Finance
- Certifications:** Supervised Machine Learning, Advanced Learning Algorithms, Unsupervised Learning

SELECTED AWARDS

- Canadian National Debate Championship - **1st speaker & team in Canada**
- Munk Debates ReSolved Global Debate Competition - **Champion, awarded \$5,000**
- JamHacks VII (Largest Canadian HS hackathon) - **First Place Overall, Best Use of GitHub**
- American Invitational Math Exam (AIME) Qualifier - **Top 5% Worldwide**